

Sai Sumanth Reddy

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience designing, developing, and delivering scalable, secure enterprise applications using Java Spring Boot, microservices architecture, and event-driven systems. Proven expertise in Apache Kafka, RESTful APIs, cloud platforms (AWS, PCF), containerization (Docker, Kubernetes), and CI/CD automation (Jenkins, Git). Strong background in full software development lifecycle from requirements analysis to production support, with demonstrated ability to solve complex technical challenges, optimize system performance, and collaborate effectively with cross-functional teams in Agile environments. Passionate about building reliable, maintainable software solutions that drive business value.

TECHNICAL SKILLS

Programming Languages: Java (Java 17, Java 11, Java 8), TypeScript, JavaScript, Python, C, C++, Go, SQL, Bash
Java Frameworks: Spring Boot, Spring MVC, Spring Batch, Spring Data JPA, Spring Security, Spring Cloud, Micronaut
Event-Driven Architecture: Apache Kafka, Kafka Streams, RabbitMQ, Event Sourcing, Asynchronous Processing
Microservices & APIs: RESTful APIs, gRPC, GraphQL, Microservices Architecture, API Gateway, Service Mesh
Frontend Technologies: JavaScript, TypeScript, React.js, Angular, Next.js, HTML5, CSS3, Responsive Design
Database Systems: PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, OpenSearch, SQL Optimization, NoSQL
Cloud Platforms: AWS (EKS, Lambda, ECS, S3, RDS, CDK), Pivotal Cloud Foundry (PCF), Azure
DevOps & CI/CD: Jenkins, Git, GitHub, GitLab, Docker, Kubernetes, Terraform, CloudFormation, Argo CD, Helm
Observability: OpenTelemetry, Prometheus, Grafana, ELK Stack, AWS CloudWatch, Application Monitoring
Security: OAuth 2.0, JWT, SAST/DAST/SCA, CIS Benchmarks, Security Best Practices, Chaos Engineering
Testing: JUnit, PyTest, Jest, Cypress, Pact, Mockito, Test-Driven Development, Integration Testing
Design Patterns: Object-Oriented Programming, SOLID Principles, Factory, Singleton, Strategy, Observer Patterns
Methodologies: Agile/Scrum, Kanban, Continuous Integration/Delivery, Performance Optimization

PROFESSIONAL EXPERIENCE

Lead Software Engineer

July 2024 – Present

Hotkey Solutions

Baltimore, MD

- Design, develop, and maintain scalable backend microservices using Java Spring Boot (Java 17) and TypeScript/Node.js, delivering secure and reliable solutions across 3 production AI-driven platforms supporting decision support, compliance tracking, and administrative workflows
- Architect and implement event-driven systems using Apache Kafka and Kafka Streams for real-time data processing, handling 50,000+ daily events with sub-second latency and 99.9% reliability across distributed microservices
- Build comprehensive RESTful APIs supporting alerts, reminders, monitoring events, structured workflows, and role-based access control (RBAC) for 3+ user roles (Admin, Provider, End User), reducing permission-related defects by 50%
- Deploy containerized applications using Docker and Kubernetes on AWS (EKS, ECS, Lambda, RDS), implementing auto-scaling, load balancing, and infrastructure-as-code using Terraform and AWS CDK
- Establish CI/CD pipelines using Jenkins, GitHub Actions, and Argo CD with automated testing, code quality checks, and deployment automation, reducing deployment time by 65% and enabling multiple daily releases
- Design and optimize PostgreSQL, DynamoDB, and Redis databases with efficient schema design, indexing strategies, and query optimization, improving query performance by 45% and supporting high-throughput operations
- Implement comprehensive observability using OpenTelemetry, Prometheus, Grafana, and ELK Stack for distributed tracing, metrics collection, and log aggregation, reducing MTTR from 4 hours to 45 minutes
- Integrate multiple applications into shared authentication ecosystem using OAuth 2.0 and JWT, improving consistency, reducing duplicate logic, and enhancing security posture across platforms
- Develop frontend components using React, Angular, and TypeScript aligned with backend APIs, supporting 5+ role-specific dashboards and real-time status tracking interfaces
- Debug and resolve critical routing, authorization, data-flow, and integration issues, reducing demo-blocking and test-blocking defects by 40-50% through systematic root cause analysis

- Collaborate with cross-functional teams including product managers, frontend developers, QA engineers, and stakeholders to gather requirements, align on technical approaches, and deliver features meeting business objectives
- Participate in Agile Scrum ceremonies including sprint planning, daily standups, retrospectives, and code reviews, consistently delivering high-quality features within sprint timelines
- Manage production support, incident management, and on-call rotations with 24/7 availability, maintaining 99.7% uptime and resolving P1 incidents within 2-hour SLA
- Create comprehensive technical documentation including API specifications, architecture diagrams, system design documents, and operational runbooks following engineering standards
- Conduct feasibility studies and evaluate alternative technical approaches for current and future requirements, recommending solutions balancing performance, scalability, and maintainability

Software Engineer Intern

January 2022 – May 2022

Wavemaker, Inc.

Hyderabad, India

- Developed full-stack features using React, TypeScript, Node.js, and Java Spring Boot, contributing to multiple production releases for internal and client-facing web applications
- Built backend REST APIs and business logic supporting user workflows, data validation, form processing, and third-party integrations with proper error handling and security measures
- Implemented responsive frontend components and UI flows using React and TypeScript aligned with product requirements, improving usability and user experience across key application screens
- Designed and optimized PostgreSQL database schemas, wrote complex SQL queries, and implemented data access layer using Spring Data JPA for efficient data retrieval and manipulation
- Debugged issues across full stack including frontend rendering, API integration, database queries, and business logic, resolving 50+ defects and improving application stability
- Wrote comprehensive unit tests using JUnit and Jest, achieving 80%+ code coverage and reducing regression defects through test-driven development practices
- Collaborated with engineers, product stakeholders, and QA in Agile environment, participating in sprint planning, daily standups, code reviews, and retrospectives
- Utilized Git version control with feature branching strategies, participated in pull request reviews, and followed coding standards and best practices
- Supported feature delivery timelines by contributing effectively across both frontend and backend layers, accelerating development velocity through versatile full-stack contributions

EDUCATION

University of Maryland, Baltimore County

Baltimore, MD

Master of Science in Information Sciences – GPA: 3.7/4.0

August 2022 – May 2024

- Relevant Coursework: Design and Analysis of Algorithms, Advanced Computer Architecture, Principles of Computer Security, Mobile Computing, Information Retrieval, Introduction to Data Science

Malla Reddy University

Hyderabad, India

Bachelor of Technology in Computer Science – GPA: 3.5/4.0

July 2018 – July 2022

- Relevant Coursework: Operating Systems, Linux Programming, Object-Oriented Programming, Database Management Systems, Big Data Analytics, Mobile Application Development

TECHNICAL PROJECTS

InsightGPT – AI-Driven Decision Support Platform

Java Spring Boot, Apache Kafka, React, PostgreSQL, AWS, Docker, Kubernetes

- Architected scalable microservices platform using Java Spring Boot and event-driven architecture with Apache Kafka for real-time alerts, reminders, monitoring events, and structured decision workflows
- Implemented comprehensive RBAC system for 3+ user roles (Admin, Provider, End User) with fine-grained permissions, improving routing accuracy and authorization enforcement by 50%
- Developed RESTful APIs supporting 5+ role-specific dashboards for tracking decisions, follow-ups, status changes, and compliance metrics with real-time updates
- Built responsive frontend using React and TypeScript with state management, form validation, and API integration for seamless user experience

- Deployed containerized services on AWS EKS with horizontal pod autoscaling, achieving 99.9% uptime and supporting 10,000+ concurrent users
- Implemented CI/CD pipeline using Jenkins and Docker, enabling automated testing, security scanning, and zero-downtime deployments
- Optimized PostgreSQL database with proper indexing, query optimization, and connection pooling, reducing average API response time to sub-200ms

Conditional Release Support Platform – Compliance Tracking System

Java Spring Boot, Spring Batch, Kafka, TypeScript, Firebase, REST APIs

- Developed compliance-focused platform using Java Spring Boot supporting daily plans, required actions, alerts, violation tracking, and status reporting across multiple concurrent cases
- Implemented event-driven architecture using Apache Kafka for asynchronous processing of compliance events, escalations, and notification workflows
- Built Spring Batch jobs for scheduled processing of missed actions, deadline monitoring, and automated violation detection across 10+ compliance workflow types
- Created role-based permission system ensuring correct visibility and data access for administrators, case managers, and compliance officers
- Designed RESTful APIs for case management, status tracking, reporting, and audit trail generation with comprehensive input validation and error handling
- Developed automated testing suite with JUnit and integration tests covering 20+ end-to-end compliance scenarios including normal flows, edge cases, and failure paths
- Reduced test-blocking and demo-blocking defects by 45% through systematic debugging, root cause analysis, and implementation of defensive programming practices

Comp Restore AI – Administrative Workflow Automation Platform

TypeScript, Node.js, React, PostgreSQL, AWS Lambda, Docker, REST APIs

- Built administrative workflow platform using TypeScript and Node.js for managing recovery, resolution, and follow-up processes across multiple concurrent workflows
- Implemented serverless backend using AWS Lambda functions with API Gateway for scalable request processing and cost-effective infrastructure
- Developed shared authentication system using OAuth 2.0 and JWT tokens, enabling single sign-on across multiple applications and reducing duplicate access control logic
- Created admin dashboards using React with real-time status updates, task management, and workflow visualization for tracking recovery progress
- Designed PostgreSQL database schema with proper normalization, constraints, and indexes supporting complex queries for reporting and analytics
- Implemented comprehensive error handling, logging, and monitoring using CloudWatch, enabling quick identification and resolution of production issues
- Validated 15+ admin workflows through iterative testing cycles, ensuring alignment with real-world use cases and stakeholder requirements